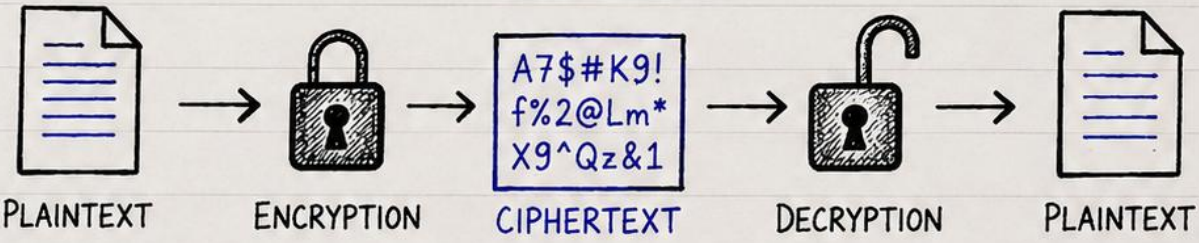


CRYPTOGRAPHY

— THE ART OF SECURE COMMUNICATION —







Protecting information.
Powering trust.



CONFIDENTIALITY
Keep it secret.



INTEGRITY
Keep it accurate.



AUTHENTICATION
Verify the identity.



NON-REPUDIATION
Prove it happened.

Secure today, Empower tomorrow.

1. Cryptocurrency Introduction + Blockchain Basics



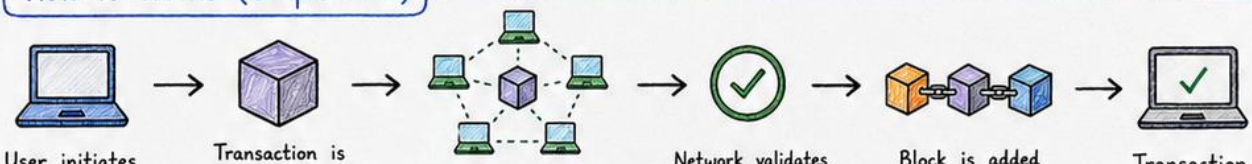
What is Cryptocurrency?

- Cryptocurrency is a digital or virtual currency that uses cryptography for security.
- It works on a technology called Blockchain.
- It is decentralized, meaning no single person or organization controls it.

Key Features

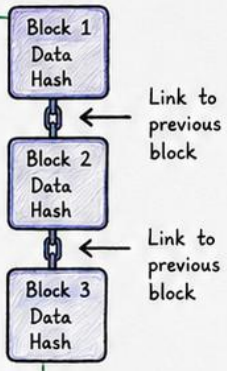
- ★ Decentralized – No central authority
- ★ Secure – Uses advanced cryptography
- ★ Transparent – All transactions are recorded on the blockchain
- ★ Immutable – Once recorded, data cannot be changed
- ★ Global – Can be sent anywhere instantly

How it Works (Simple Flow)



Blockchain Basics

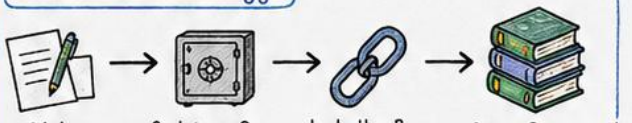
- A Blockchain is a digital ledger that stores data in blocks.
- Each block contains:
  - ✓ Transaction Data
  - ✓ Timestamp
  - ✓ Previous Block Hash
  - ✓ Unique Hash
- Blocks are linked in a chain, making it tamper-proof.



Types of Cryptocurrencies

- Bitcoin (BTC) – First and most popular cryptocurrency.
- Altcoins – All cryptocurrencies other than Bitcoin (e.g., ETH, BNB).
- Stablecoins – Pegged to fiat currency like USD to reduce volatility (e.g., USDT, USDC).
- Tokens – Built on top of existing blockchains (e.g., ERC-20 tokens).

Real World Analogy



Why is it Important?

- ✓ Enables secure peer-to-peer transactions.
- ✓ Reduces need for intermediaries (banks).
- ✓ Provides financial freedom and inclusion.
- ✓ Basis for future technologies like DeFi, NFTs, Web3 and more.



Cryptocurrency is not just about money, it's about a new way to build trust in the digital world.





2. BITCOIN + HOW BITCOIN WORKS

What is Bitcoin?

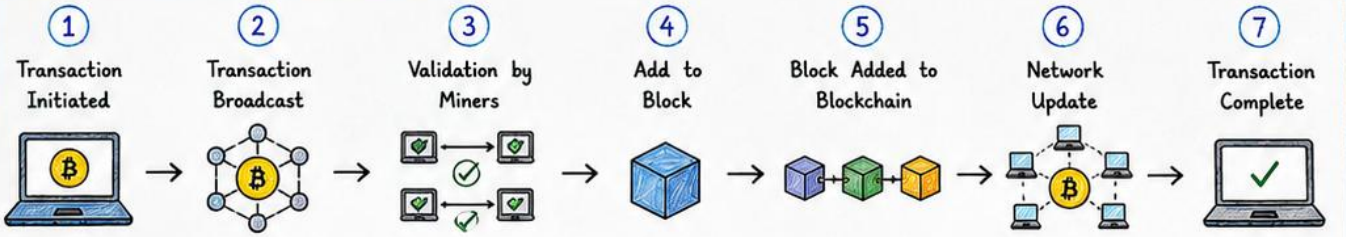
- Bitcoin (BTC) is the first and most popular cryptocurrency.
- It was created in 2009 by an unknown person or group named Satoshi Nakamoto.
- Bitcoin is a decentralized digital currency that allows peer-to-peer transactions without a middleman.
- Total supply is limited to 21 million BTC.

Key Features of Bitcoin

- ★ Decentralized – No central authority or bank control.
- ★ Limited Supply – Only 21 million BTC will ever exist.
- ★ Secure – Uses cryptography and blockchain technology.
- ★ Transparent – All transactions are public and verifiable.
- ★ Immutable – Once recorded, transactions cannot be changed.
- ★ Permissionless – Anyone can send, receive and store Bitcoin.

Bitcoin is digital gold – scarce, secure and decentralized. It is mainly used as a store of value and medium of exchange.

How Bitcoin Works (Step-by-Step)



Bitcoin Components

- Transaction – Transfer of BTC between addresses.
- Block – A container that stores multiple transactions.
- Blockchain – A chain of blocks linked together.
- Miner – A participant who validates transactions and adds blocks.
- Hash – A unique code that identifies each block.
- Node – Any computer connected to the Bitcoin network.

Example

A sends 0.5 BTC to B
↓
Miners validate it
↓
Added to Block #1250
↓
Block #1250 added to blockchain
↓
B receives 0.5 BTC

Bitcoin Facts

- Launched: January 3, 2009
- Creator: Satoshi Nakamoto
- Max Supply: 21,000,000 BTC
- Halving: Occurs every ~4 years
- Current Use: Store of value, payments, investment

Advantages of Bitcoin

- ✓ Decentralized and not controlled by any government.
- ✓ Secure and transparent.
- ✓ Low transaction fees compared to banks.
- ✓ Can be sent globally in minutes.
- ✓ Supply is limited (deflationary asset).

Limitations of Bitcoin

- ✗ Price is volatile.
- ✗ Transactions are not instant (takes time for confirmation).
- ✗ Energy intensive (mining requires high power).
- ✗ Limited scalability (7 transactions per second approx).

Did You Know?

The first real-world purchase using Bitcoin was 10,000 BTC for two pizzas on May 22, 2010 (Bitcoin Pizza Day?).

Bitcoin is the foundation of the crypto world.
Understand Bitcoin, and you understand cryptocurrency.

3. ETHEREUM + SMART CONTRACTS

What is Ethereum?

- Ethereum is a decentralized, open-source blockchain platform.
- It was proposed by Vitalik Buterin in 2013 and launched in 2015.
- Ethereum is not just a cryptocurrency (ETH), it is a platform for building decentralized applications (dApps).
- It introduced the concept of Smart Contracts.

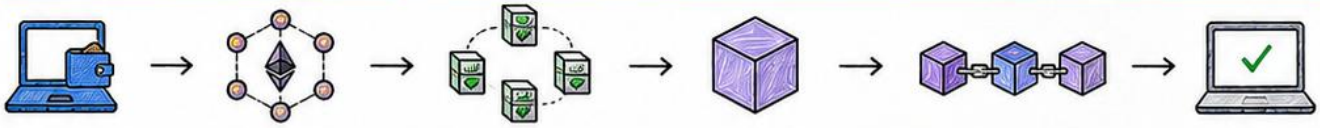
Key Features of Ethereum

- ✓ Decentralized platform
- ✓ Supports Smart Contracts and dApps
- ✓ Ethereum Virtual Machine (EVM)
- ✓ Uses Ether (ETH) as native cryptocurrency
- ✓ Large developer ecosystem
- ✓ Upgradable and flexible

Ethereum vs Bitcoin

Feature	Bitcoin	Ethereum
Purpose	Digital Currency	Decentralized Platform
Launched	2009	2015
Blockchain Use	Payments	Smart Contracts & dApps
Native Coin	BTC	ETH
Scripting Capability	Limited	Advanced (Turing Complete)

How Ethereum Works (High Level)



1. User creates a transaction or runs a dApp.
2. The transaction is broadcast to the Ethereum network.
3. Validators (nodes) validate the transaction using consensus.
4. Once validated, the transaction is added to a block.
5. The block is added to the blockchain permanently.
6. The transaction is complete and visible to everyone.

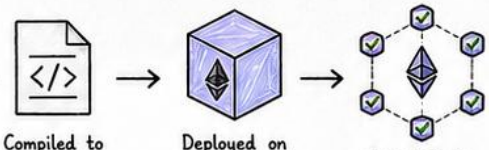
What is a Smart Contract?

- A smart contract is a self-executing program stored on the Ethereum blockchain.
- It automatically executes when predefined conditions are met.
- It runs exactly as coded and cannot be changed once deployed.
- Smart contracts enable trustless, transparent and intermediary-free transactions.

Smart Contract Example (Simple)

Code (Solidity)

```
pragma solidity ^0.8.0;
contract Hello {
   string public msg =
     "Hello Ethereum!";
   function setMsg(string
   memory _msg) public {
      msg = _msg;
   }
}
```



Compiled to Bytecode → Deployed on Ethereum Blockchain → Executed by EVM on all nodes

Ethereum Virtual Machine (EVM)

- EVM is the runtime environment for smart contracts.
- It executes smart contract code on every node in the network.
- Ensures all nodes get the same result.
- Provides security, isolation and consistency.



Use Cases of Ethereum

- ★ Decentralized Finance (DeFi)
- ★ NFTs and Digital Collectibles
- ★ Decentralized Applications (dApps)
- ★ Gaming and Metaverse
- ★ DAOs (Decentralized Autonomous Organizations)



Important Terms

- ETH (Ether) – Native cryptocurrency of Ethereum
- dApp – Decentralized Application
- Smart Contract – Self-executing contract
- Gas – Fee paid to execute transactions or smart contracts
- Address – Unique identifier on Ethereum



Benefits of Smart Contracts

- ✓ Automated – No need for intermediaries
- ✓ Trustless – No need to trust a third party
- ✓ Transparent – Code and transactions are public
- ✓ Secure – Runs on blockchain, hard to tamper
- ✓ Efficient – Saves time and reduces cost

Limitations

- ✗ Bug in code cannot be changed easily
- ✗ High Gas fees during network congestion
- ✗ Scalability issues (Ethereum 1.0 had limits)
- ✗ Smart contracts depend on external data (oracle risk)
- ✗ Complex to develop and audit

Did You Know?

Ethereum is more than a currency – it is a world computer that can run any decentralized application!





Ethereum brought programmability to blockchain.
Smart Contracts are the building blocks of Web3!



4. BLOCKCHAIN ARCHITECTURE + TRANSACTIONS

What is Blockchain Architecture?

- Blockchain architecture is the structure that defines how data is organized, stored, validated and shared across the network.
- It ensures decentralization, security, transparency and immutability.

Key Characteristics

- ✓ Decentralized
- ✓ Distributed Ledger
- ✓ Immutable
- ✓ Transparent
- ✓ Secure (Cryptography)
- ✓ Consensus Driven

Types of Blockchain Architecture

Public Blockchain
Open to anyone. Fully decentralized. (Example: Bitcoin, Ethereum)

Private Blockchain
Restricted access. Controlled by one organization.

Consortium (Federated) Blockchain
Controlled by a group of organizations. Semi-decentralized.

Blockchain Architecture (High Level)

Application Layer	DApps, Smart Contracts, User Interfaces
Network Layer	P2P Network, Node Discovery, Data Propagation
Consensus Layer	Validation of Transactions (PoW, PoS, etc.)
Data Layer	Blocks, Transactions, Merkle Trees, Hashing
Infrastructure Layer	Nodes, Storage, Compute, Communication

Transactions in Blockchain

A transaction is a transfer of value/data between parties on the blockchain.

Transaction Flow (Step-by-Step)

1. **Transaction Initiated**
   A sends data/crypto to B.
2. **Transaction Broadcasted**
   The transaction is broadcasted to all nodes in the network.
3. **Validation by Nodes**
   Nodes validate the transaction based on consensus rules.
4. **Transaction Added to Block**
   Once validated, the transaction is added to a new block.
5. **Block Added to Blockchain**
   The block is added to the blockchain and becomes permanent.
6. **Transaction Complete**
   B receives the assets. Transaction is complete and recorded.

Anatomy of a Block

Block Header	
Version	
Previous Block Hash	Links to the previous block
Merkle Root	
Timestamp	Hash of all transactions in the block
Difficulty	
Nonce	
Transaction Data	
Tx 1	List of all transactions
Tx 2	
Tx 3	
...	
Tx n	

Blockchain Structure

Block n-1	Block n	Block n+1
Header	Header	Header
Txs	Txs	Txs
...	...	...

Each block contains the hash of the previous block, creating a chain.

Immutability:
If one block is changed, all subsequent blocks become invalid.

Consensus Mechanisms (Examples)

Proof of Work (PoW)
- Miners solve complex mathematical puzzles.
- First to solve gets to add the block.
- Example: Bitcoin

Proof of Stake (PoS)
- Validators are chosen based on the stake they hold.
- More energy efficient than PoW.
- Example: Ethereum 2.0

Key Takeaways

- ✓ Blockchain is a decentralized and distributed ledger.
- ✓ Transactions are verified by the network before being added to the blockchain.
- ✓ Once recorded, data cannot be altered.
- ✓ Transparency + Security + Immutability = Trust

Did You Know?

Bitcoin's blockchain has been running non-stop since January 3, 2009!

Blockchain eliminates the need for intermediaries and builds trust through code and cryptography.

5. WALLETS + PUBLIC & PRIVATE KEYS



What is a Crypto Wallet?

- A crypto wallet is a tool that allows you to store, send and receive cryptocurrencies.
- Wallets do not store coins directly. They store your keys which give you access to the blockchain.
- You are your own bank!



Types of Wallets

1. Hot Wallets (Connected to Internet)



- Easy to use
- Suitable for small amounts

Examples: MetaMask, Trust Wallet, Coinbase Wallet

2. Cold Wallets (Offline)



- More secure
- Suitable for long-term storage

Examples: Ledger, Trezor (Paper Wallets)

Wallet Types Comparison

Hot Wallet	Cold Wallet
Connected to Internet	Offline (Private)
Convenient & Fast	More Secure
Higher Risk (Hacks, Malware)	Lower Risk
For Daily Use (Small Funds)	For Long-Term Storage (Large Funds)

Public Key vs Private Key

Public Key (Share it)

- Derived from your private key
- Like your bank account number
- Can be shared with anyone
- Used to receive crypto

Example Address:

`1A1zP1eP5QGefi2DMPTfTL5SLmv7DivfNa`

(This is a Bitcoin address)



Two keys, One Wallet

Private Key (Keep it Secret!)

- A secret alphanumeric code
- Like your bank PIN / password
- Must be kept private
- Used to sign transactions and access your funds

Example Private Key:

`5Kb8kLf9t12eWgtjMV3H7B8D5K98abc123xyz...`

(Never share this with anyone!)

Important Rule

Whoever has your private key, has control over your funds.

NOT your keys, NOT your coins.

How Keys Work? (Simplified)

Private Key — A random secret number is generated.

→

Public Key — Derived from the private key using mathematical algorithms.

→

Wallet Address — Derived from the public key. This is what others use to send you crypto.

→

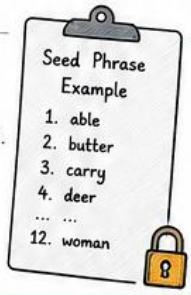
Transaction — You sign the transaction with your private key to prove ownership.

→

Blockchain — Network verifies using your public key and records the transaction. Funds are transferred!

Wallet Security Best Practices

- ✓ Always keep your private key secret.
- ✓ Use strong passwords and 2FA.
- ✓ Backup your seed phrase (write it down and store safely).
- ✓ Never share your seed phrase or private key.
- ✓ Beware of phishing websites and fake apps.
- ✓ Use hardware wallets for large amounts.
- ✓ Keep your devices and software updated.



What is a Seed Phrase?

- A seed phrase (12/24 words) is a human-readable backup of your wallet.
- It can regenerate all your keys and restore your wallet.
- Treat it like your life savings!



Example: Sending Crypto

1. You want to send crypto to someone.
2. You enter the recipient's public address.
3. You sign the transaction with your private key.
4. Network verifies the signature using your public key.
5. Transaction is added to blockchain and completed.



Quick Summary

- ★ Wallets store your keys, not coins.
- ★ Public Key = Share it (Receive)
- ★ Private Key = Keep it secret (Access)
- ★ Seed Phrase = Backup (Recovery)
- ★ Security = Protect your keys, Protect your wealth.



Remember: Blockchain is public, but your keys are private.
Secure your keys, secure your future.



6. EXCHANGES + BUYING & SELLING CRYPTO



What is a Crypto Exchange?

- A crypto exchange is a platform where you can buy, sell and trade cryptocurrencies.
- It acts as a marketplace connecting buyers and sellers.
- Exchanges may be centralized (CEX) or decentralized (DEX).



Types of Exchanges

1. Centralized Exchange (CEX)

- ✓ Run by a company
- ✓ Easy to use
- ✓ High liquidity
- ✓ Requires KYC

Examples: Binance, Coinbase, Kraken, KuCoin



2. Decentralized Exchange (DEX)

- ✓ No central authority
- ✓ You control your funds
- ✓ More privacy
- ✓ Lower liquidity

Examples: Uniswap, PancakeSwap, dYdX, 1inch



CEX vs DEX (Quick Comparison)

Feature	CEX	DEX
Control	Platform	User
KYC	Required	Not Required
Security	Depends on Exchange	You control your funds
Liquidity	High	Lower
Ease of Use	Easy	Complex
Examples	Binance, Coinbase	Uniswap, PancakeSwap

How to Buy Crypto (Step-by-Step on a CEX)

1. **Create Account** — Sign up with your email/phone and secure your account with 2FA.
2. **Complete KYC** — Submit identity documents as per exchange policy.
3. **Deposit Funds** — Add money via bank transfer, card, UPI or other payment methods.
4. **Choose Trading Pair** — Select the crypto you want to buy, e.g., BTC/USDT.
5. **Place Order** — Choose order type:
   - Market Order (buy at current price)
   - Limit Order (set your price)
6. **Crypto in Wallet** — After purchase, crypto will be in your exchange wallet.
7. **Secure (Optional)** — Transfer to your personal wallet for maximum security.

How to Sell Crypto (Step-by-Step on a CEX)

1. **Go to Wallet** — Open your spot wallet and select the crypto you want to sell.
2. **Choose Trading Pair** — Select the pair, e.g., BTC/USDT.
3. **Place Sell Order** — Choose Market Order or Limit Order and place the sell order.
4. **Funds in Account** — Once order is filled, funds (e.g., USDT) will be added to your account.
5. **Withdraw to Bank** — Withdraw to your bank via UPI, IMPS, NEFT or other methods.
6. **Verify & Confirm** — Complete any required verification by the exchange.
7. **Funds Received** — Your money will reflect in your bank account.

Popular Payment Methods

- Bank Transfer (NEFT/IMPS)
- Debit/Credit Card
- UPI
- P2P Trading

Note: Methods depend on your country and exchange.

Order Types

- Market Order – Buy/Sell instantly at current market price.
- Limit Order – Set your own price to buy/sell.
- Stop-Limit Order – Triggers when price reaches a certain level.

Fees You May Pay

- Deposit Fee – Usually low or zero
- Trading Fee – 0.1% to 0.5% (varies)
- Withdrawal Fee – Depends on coin & network
- Network Fee (Gas) – For on-chain transactions

Safety Tips

- ✓ Use trusted exchanges
- ✓ Enable 2FA
- ✓ Never share your passwords
- ✓ Withdraw large holdings to your personal wallet
- ✓ Beware of phishing scams

CEX Examples

Binance · Coinbase · Kraken · KuCoin

Did You Know?

The world's largest exchange by trading volume is Binance.

DEX Examples

Uniswap · PancakeSwap · dYdX · 1inch

Exchanges make crypto accessible, but smart users stay safe.
Learn, compare and choose the platform that suits you best!

7. ALTCOINS + STABLECOINS

What are Altcoins?

- Altcoin = Any cryptocurrency other than Bitcoin (BTC).
- "Alt" stands for Alternative.
- Thousands of altcoins exist with different use cases, technologies and goals.
- They aim to improve, innovate or solve specific problems that Bitcoin doesn't.

Popular Altcoins & Their Use Cases

Altcoin	Ticker	Use Case / Purpose
Ethereum	ETH	Smart contracts, dApps, DeFi, NFTs
Binance Coin	BNB	Powers Binance ecosystem, pay trading fees, utilities
Solana	SOL	High-speed blockchain for dApps and payments
Cardano	ADA	Research-driven blockchain for security and scalability
Polkadot	DOT	Connects multiple blockchains (Interoperability)
Chainlink	LINK	Decentralized oracle network (real-world data for smart contracts)
Polygon	MATIC	Layer-2 scaling solution for Ethereum, low fees

Key Points About Altcoins

- ✓ Higher risk but higher reward potential.
- ✓ Many altcoins are volatile.
- ✓ Do your own research before investing.
- ✓ Each altcoin has a unique purpose and technology.

What are Stablecoins?

- Stablecoins are cryptocurrencies designed to maintain a stable value.
- Their value is pegged (linked) to a stable asset like fiat currency (USD), gold, or other assets.
- They reduce volatility in the crypto market.

Types of Stablecoins (By Collateral)

1. Fiat-Collateralized Stablecoins

Backed by fiat currency (reserves in bank).

Example: USDT, USDC

2. Crypto-Collateralized Stablecoins

Backed by other crypto assets (over-collateralized).

Example: DAI

3. Algorithmic Stablecoins

Not backed by reserves. Use algorithms to maintain price stability.

Example: UST (earlier)

Popular Stablecoins

Stablecoin	Ticker	Pegged To
Tether	USDT	1 USD
USD Coin	USDC	1 USD
Binance USD	BUSD	1 USD
Dai	DAI	1 USD (Over-Collateralized)
TrueUSD	TUSD	1 USD

Why Use Stablecoins?

- Store value without volatility
- Easy trading between cryptocurrencies
- Used in DeFi for lending, borrowing, liquidity pools
- Global payments and remittances
- Used for purchases and services

Altcoins vs Stablecoins

Feature	Altcoins	Stablecoins
Price Volatility	High	Low (Stable)
Purpose	Innovation, Utility, Growth	Stability, Store of Value, Transactions
Backing	Usually not backed	Backed by fiat, crypto or algorithm
Risk Level	Higher	Lower
Use Case	Trading, Investing, DeFi, dApps	Payments, Trading, DeFi, Savings

Examples in Real Life

Altcoins

ETH is used for smart contracts.
SOL is used for fast dApps.
LINK provides data to real-world applications.

Stablecoins

USDT/USDC are used to park funds during market volatility.
Businesses accept stablecoins for payments.

Things to Remember

- ✓ Altcoins are not all the same. Understand their technology and team.
- ✓ Stablecoins are safer but still not 100% risk-free (e.g., depeg risk).
- ✓ Use stablecoins in DeFi to earn interest or provide liquidity.
- ✓ Diversify and manage risk wisely.

Risks to Know

- ⚠ Altcoins can be highly volatile.
- ⚠ Some altcoins may fail.
- ⚠ Stablecoins can lose peg (e.g., UST collapse).
- ⚠ Regulatory risks for stablecoins and exchanges.

Did You Know?

As of 2024, there are 20,000+ altcoins in the crypto market!

Top Stablecoins by Market Cap (Example)

1. Tether (USDT)
2. USD Coin (USDC)
3. Binance USD (BUSD)
4. Dai (DAI)

Pro Tip

Use stablecoins to protect your profits during market downturns.

Altcoins bring innovation. Stablecoins bring stability.
Use both wisely to grow and protect your wealth in crypto!

8. TOKENS + NFTs

What is a Token?

- A token is a digital asset built on an existing blockchain.
- Unlike coins (like BTC, ETH), tokens do not have their own blockchain.
- They represent a value or utility within a project or ecosystem.
- Tokens are created using smart contracts.

Types of Tokens

Type	Description	Examples
Utility Tokens	Provide access to a product or service within a platform.	BAT (Basic Attention Token), LINK (Chainlink)
Security Tokens	Represent ownership in real-world assets (securities).	tZERO (TZROP), RealT
Governance Tokens	Give holders the right to vote on protocol decisions.	UNI (Uniswap), AAVE, COMP
Payment Tokens	Used as a medium of exchange.	USDT, USDC (Before being stablecoins)
DeFi Tokens	Used in DeFi protocols for incentives, yield, liquidity, etc.	SUSHI, YFI, CRV

What are NFTs?

- NFT stands for Non-Fungible Token.
- NFTs are unique digital assets that cannot be replaced with something else.
- Each NFT has distinct information and ownership recorded on the blockchain.
- NFTs are commonly used for digital art, collectibles, gaming items, domain names and more.

Token Standards

Tokens are created using standards on blockchains (mostly Ethereum).

Standard	Type	Description	Example
ERC-20	Fungible	Most common token standard. Tokens are identical and interchangeable.	USDT, UNI, LINK
ERC-721	Non-Fungible	Standard for NFTs. Each token is unique.	CryptoPunks, Bored Ape
ERC-1155	Multi-Token	Supports both fungible and non-fungible tokens in one smart contract.	Axie Infinity items

NFT Standards

NFTs follow specific standards to ensure uniqueness and ownership.

Standard	Description
ERC-721	The original NFT standard. Each token has a unique ID and metadata.
ERC-1155	More efficient standard. Supports multiple NFT types and fungible tokens in a single contract.

Example of an NFT

Digital Asset (Art, Music, etc.) → Metadata (Unique Info) → Stored on Blockchain → Unique NFT with Proven Ownership

Coins vs Tokens

Feature	Coins (e.g., BTC, ETH)	Tokens (e.g., USDT, UNI)
Blockchain	Have their own blockchain	Built on existing blockchain
Purpose	Main currency of the network	Built for specific use cases
Examples	Bitcoin, Ethereum	USDT, LINK, UNI
Creation	Created during blockchain launch	Created anytime using smart contracts

Popular NFT Use Cases

Digital Art · Gaming Items · Event Tickets · Domains (Web3) · Music & Videos · Collectibles

How Tokens Are Used?

- ✓ Access to platform features
- ✓ Pay fees on the platform
- ✓ Earn rewards and incentives
- ✓ Governance and voting rights
- ✓ Staking and yield farming
- ✓ Represent assets or ownership

Examples in Real Life

- ★ UNI → Vote on Uniswap decisions
- ★ LINK → Data feeds for smart contracts
- ★ USDT → Used for trading and payments
- ★ SUSHI → Rewards for liquidity providers

Why NFTs are Valuable?

- ✓ Unique and one-of-a-kind
- ✓ Verifiable ownership on blockchain
- ✓ Cannot be copied or replaced
- ✓ Creators can earn royalties on resale
- ✓ Global market and transparency

Things to Remember

- ★ Tokens can be fungible or non-fungible.
- ★ NFTs are a type of token (ERC-721 or ERC-1155).
- ★ Value comes from utility, community and adoption.

Famous Token Examples

USDT (Utility) · LINK (Utility) · UNI (Governance) · DAI (Stablecoin) · AAVE (Governance)

Famous NFT Collections

CryptoPunks · Bored Ape Yacht Club · Azuki · Cool Cats · Doodles

Tokens power ecosystems. | NFTs represent uniqueness.
Both are building the future of Web3!

9. DeFi + Decentralized Exchanges (DEX)

What is DeFi?

- DeFi (Decentralized Finance) refers to financial services built on blockchain, without intermediaries like banks.
- It uses smart contracts to provide lending, borrowing, trading, earning interest and more.
- Open, permissionless, transparent and accessible to anyone with an internet connection.

Key Features of DeFi

Decentralized	No central authority or intermediary.
Permissionless	Anyone can access and use DeFi protocols.
Transparent	All transactions are on-chain and publicly verifiable.
Composable	DeFi protocols can integrate like money legos.
Global & Open	Available 24/7 to anyone, anywhere.

Common DeFi Use Cases

Lending & Borrowing
Lend your crypto and earn interest or borrow by providing collateral.

Decentralized Trading
Trade tokens directly from your wallet on DEXs.

Yield Farming
Provide liquidity and earn rewards in tokens.

Staking
Lock your tokens to support the network and earn rewards.

Stablecoins
Used for stability, trading and savings in DeFi.

DeFi Ecosystem (How it Works)

You interact with DeFi using a crypto wallet.

You → Crypto Wallet (MetaMask, etc.) → DeFi Protocols (Smart Contracts) → Financial Services (Lend, Borrow, Swap, Earn) → You Earn (Yield / Rewards)

You stay in control of your funds. No bank. No middleman.

Popular DeFi Protocols

Protocol	Category	What it does
Aave	Lending	Lend & borrow crypto assets
Compound	Lending	Earn interest or borrow assets
Uniswap	DEX	Swap tokens on Ethereum
Curve	DEX (Stablecoin)	Efficient stablecoin trading
Yearn Finance	Yield Optimizer	Maximize yield across protocols
MakerDAO	Stablecoin	Issuer of DAI stablecoin

What is a DEX?

- DEX (Decentralized Exchange) allows users to trade crypto directly from their wallets.
- No central authority holds your funds.
- Trades happen on smart contracts using automated liquidity.
- You always remain in control of your assets.

Centralized Exchange (CEX) — You don't own your coins — You own your coins — Decentralized Exchange (DEX)

How Decentralized Exchanges (DEX) Work

1. **Connect Wallet** — You connect your wallet (e.g., MetaMask).
2. **Choose Token Pair** — Select the tokens you want to trade, e.g., ETH / USDC.
3. **Provide Liquidity (Behind the scenes)** — Liquidity pools (from other users) are used to execute the trade.
4. **Smart Contract Executes Trade** — The trade is executed automatically by the smart contract.
6. **Transaction on Blockchain** — The transaction is recorded on the blockchain.

CEX vs DEX (Quick Comparison)

Feature	CEX (Centralized)	DEX (Decentralized)
Control of Funds	Exchange holds your funds	You hold your funds
KYC Required	Yes, in most cases	No, permissionless
Privacy	Lower	Higher
Security Risk	Higher (hacks, exit scams)	Lower (smart contract risks)
Ease of Use	Easy for beginners	Slightly complex
Examples	Binance, Coinbase, Kraken	Uniswap, SushiSwap, dYdX

Popular Decentralized Exchanges

Uniswap (Ethereum) · SushiSwap (Ethereum) · PancakeSwap (BNB Chain) · dYdX (Derivatives) · 1inch (Aggregator) · CowSwap (Aggregator)

AMM (Automated Market Maker)

Most DEXs use AMM instead of order books. Prices are determined by liquidity pools using algorithms (e.g., $x * y = k$).

$$x \cdot y = k$$

Slippage

Due to price movement in pools, you may get slightly less or more tokens than expected. Higher slippage = higher impact.

Gas Fees

You pay gas fees (network fees) for every transaction on DEXs, depending on the blockchain you use.

Risks to Know

- Smart contract bugs
- Impermanent loss in liquidity providing
- High gas fees (network congestion)
- Scams & fake tokens.
- Market volatility

Tips for Safe DeFi & DEX Usage

- Always use trusted protocols and official websites.
- Double-check token contracts and links.
- Start with small amounts to test.
- Use hardware wallets for large holdings.
- Keep your private keys and seed phrase safe.

Key Takeaway

DeFi and DEXs are building the future of finance – open, global and user-owned. Understand how they work, manage risks and explore the endless possibilities!

DeFi gives you financial freedom. DEXs give you true ownership.
Be your own bank. Learn. Explore. Grow.

10. MINING + PROOF OF WORK (PoW)

What is Mining?

- Mining is the process of validating transactions and adding them to the blockchain.
- Miners use powerful computers to solve complex mathematical problems.
- In return, miners are rewarded with block rewards and transaction fees.

What is Proof of Work (PoW)?

- Proof of Work is a consensus mechanism used by blockchains like Bitcoin.
- Miners (nodes) compete to solve a difficult puzzle.
- The first miner to solve it gets the right to add the next block and earn rewards.
- It prevents double spending and keeps the network secure.

Solve Puzzle → Add Block & Get Reward

Why is PoW Important?

- ✓ Secures the network from attacks.
- ✓ Makes it costly to cheat or change data.
- ✓ Decentralized – no central authority.
- ✓ Has been battle-tested (Bitcoin since 2009).

How Mining & Proof of Work Works (Step-by-Step)

1. **Pending Transactions** – Users send transactions. They are grouped together into a block (called block candidate).
2. **Block Candidate Created** – The block contains transactions, timestamp, previous block hash and a nonce.
3. **Miners Compete** – Miners use powerful hardware to solve the puzzle.
4. **Solve the Puzzle** – Miners try different nonces until the block hash is less than the target value.
5. **Proof Found!** – The first miner to find the valid hash broadcasts the block to the network.
6. **Block Verified** – Other nodes verify the block and transactions. If valid, they accept it.
7. **Block Added & Reward Earned** – The block is added to the blockchain and the miner gets reward + transaction fees.

The Mining Puzzle (Simplified)

Miners must find a nonce such that:

`SHA256(Block Data + Nonce) < Target`

Where,

- SHA256 = Hash function
- Block Data = All block information
- Target = Very small number set by the

Example (Simplified)

Nonce	Hash (Example)	Valid?
1001	8f3a...7bcd	✗
1002	b29e...91af	✗
1003	0000a1b2c3...	✓ (Less than target)
1004	f91c...aa12	✗
...	...	...

Mining Hardware

More power = More chances to win!

- CPU Mining – Very slow (not used now)
- GPU Mining – Faster than CPU
- ASIC Mining – Special machines built only for mining (Most efficient)

PoW in Bitcoin

- Bitcoin uses SHA-256 algorithm.
- Target adjusts every 2016 blocks to maintain ~10 minutes block time.
- Higher difficulty = More security.
- Mining consumes lots of electricity due to high computations.

Pros & Cons of Proof of Work

☑ Pros

- Highly secure
- Decentralized
- Sybil-attack resistant
- Proven track record (Bitcoin)

✗ Cons

- High energy consumption
- Slower transactions
- Expensive hardware
- Not very eco-friendly

PoW vs Other Consensus Mechanisms

Mechanism	How it Works	Energy Use	Examples
PoW	Solve complex puzzles	High ▲	Bitcoin, Litecoin
PoS	Stake coins to get chosen	Low ▼	Ethereum 2.0, Cardano
DPoS	Vote for delegates	Very Low ▼	EOS, TRON
PoA	Approved validators	Very Low ▼	VeChain, BSC

Fun Fact

The Bitcoin network has done more than exabytes of computations since 2009!

One of the most secure networks in the world!

Mining Rewards (Bitcoin Example)

Block Reward (halves every ~4 years) + Transaction Fees (Included in the block) = Total Reward for Miner

Key Takeaway

- ✓ Mining validates transactions.
- ✓ Proof of Work keeps the network secure.
- ✓ Miners earn rewards for their work.
- ✓ High security comes at the cost of high energy.

Mining built the foundation. Proof of Work secured the trust.
Without PoW, blockchains can't be decentralized and tamper-proof!

11. PROOF OF STAKE + CONSENSUS MECHANISMS

What is Proof of Stake (PoS)?

- Proof of Stake is a consensus mechanism where validators are chosen to create blocks based on the number of coins they "stake" as collateral.
- It is an alternative to Proof of Work (PoW) that saves energy and is more scalable.
- The more you stake, the higher your chance of being selected to validate the next block and earn rewards.



Stake (Crypto) → Selected as Validator → Earn Rewards

How Proof of Stake Works (Step-by-Step)

1. **Lock / Stake Your Coins** — Validators lock their coins in the network as a stake.
2. **Validator Selection** — The protocol randomly selects a validator based on their stake, not computing power.
3. **Propose Block** — The selected validator proposes a new block of transactions.
4. **Validate by Other Validators** — Other validators verify the block and the transactions are valid.
6. **Block Added to Blockchain** — Once verified, the block is added to the blockchain.

In PoS, you don't compete with hardware, you participate with stake. It's secure, efficient and eco-friendly!

PoS vs PoW (Quick Comparison)

Feature	PoW	PoS
How it works	Miners solve complex puzzles	Validators stake coins to validate
Energy Use	Very High (Consumes a lot of electricity)	Very Low (Energy efficient)
Security	High (Proven over years)	High (Economic incentives + slashing)
Speed	Slower	Faster
Scalability	Limited	High
Cost to Participate	High (Need expensive hardware)	Low (Just need coins to stake)
Examples	Bitcoin	Ethereum 2.0, Cardano, Solana, Polkadot

Popular Consensus Mechanisms

Mechanism	How it Works	Key Features	Advantages	Disadvantages	Used By (Examples)
Proof of Work (PoW)	Miners solve complex mathematical puzzles. First to solve gets to add the block.	• High security • Decentralized	• Very secure • Proven track record	• High energy consumption • Slow transactions	Bitcoin, Litecoin, Dogecoin
Proof of Stake (PoS)	Validators are chosen to create blocks based on the number of coins they stake.	• Energy efficient • Scalable	• Low energy use • Faster & scalable	• "Rich get richer" risk • Needs economic model incentives	Ethereum 2.0, Cardano, Solana, Polkadot
Delegated Proof of Stake (DPoS)	Token holders vote for a small number of delegates (witnesses) to produce blocks.	• Voting based • Very fast	• High throughput • User governance	• More centralized • Delegate collusion risk	EOS, TRON, Lisk
Proof of Authority (PoA)	Only approved validators (known entities) can create blocks.	• Identity based • Permissioned	• Very fast • Low energy	• Less decentralized • Trust required	VeChain, BSC (earlier version)
Proof of History (PoH)	Uses cryptographic timestamps to prove the passage of time between transactions.	• Time ordered • High speed	• Very high performance • Scalable	• Depends on central time source	Solana
Practical Byzantine Fault Tolerance (PBFT)	Validators communicate and reach agreement even if some are malicious or offline.	• Fault tolerant • Finality	• Fast finality • Good for permissioned networks	• Not ideal for large public networks	Hyperledger Fabric

Ethereum: From PoW to PoS

PoW (Ethereum 1.0) → PoS (Ethereum 2.0)

- ✓ The Merge (Sep 2022) moved Ethereum from PoW to PoS.
- ✓ ~99.95% less energy consumption.
- ✓ Staking ETH helps secure the network.
- ✓ Validators earn rewards for participation.

Benefits of PoS & Modern Consensus

- ☑ Energy efficient and eco-friendly
- ☑ Faster block times and higher throughput
- ☑ Lower barriers to participate
- ☑ Economic security through staking
- ☑ Better scalability for future adoption

Challenges & Risks

- ⚠ Centralization risk (large holders control more)
- ⚠ Slashing penalty for malicious behavior
- ⚠ Complex economic models
- ⚠ Long-term security is still evolving

How Validators Earn Rewards in PoS

Stake Coins → Validate Blocks → Earn Rewards → Keep Network Secure

Rewards = Block Rewards + Transaction Fees

Real-World Use Cases

DeFi Platforms · Payments · Supply Chain · Gaming & NFTs · Enterprise Blockchains

Key Takeaway

Consensus mechanisms are the backbone of blockchain networks.

PoS is the future – secure, scalable and sustainable.

Did You Know?

By 2024, more than 60% of the top blockchains by market cap use PoS or PoS-based mechanisms!

Top PoS-Based Blockchains (by Market Cap)

Ethereum (ETH) · Cardano (ADA) · Solana (SOL) · Avalanche (AVAX) · Polkadot (DOT)

Pro Tip

Stake wisely, diversify, and always understand the tokenomics before participating in any PoS network.

Different mechanisms, one goal: DECENTRALIZED TRUST!

12. CRYPTO SECURITY + COMMON SCAMS

Why Security Matters?

- Crypto is decentralized.
  No bank = No support center.
- You are your own bank.
  You own your funds and your responsibility.
- Transactions are irreversible.
  Once it's gone, it's gone.

> "Not your keys, not your coins."

Crypto Security Best Practices

- **Use Strong Wallets** — Use reputable wallets (Ledger, Trezor, Trust Wallet, MetaMask). ✓
- **Secure Your Private Keys** — Never share your private key or seed phrase with anyone. ✓
- **Backup Your Seed Phrase** — Write it down on paper and store it in a safe place. Never store online. ✓
- **Enable 2FA Everywhere** — Use 2FA for exchanges, email, and all important accounts. ✓
- **Verify Before You Click** — Always double-check URLs, addresses and contracts. ✓
- **Keep Software Updated** — Update your wallet, browser and OS regularly. ✓

Common Security Threats

- **Phishing** — Fake websites or emails steal your login or seed phrase.
- **Malware** — Malicious software can steal your keys or clipboard data.
- **Fake Apps & Extensions** — Fake wallet apps or browser extensions steal your funds.
- **Public Wi-Fi Risks** — Avoid accessing wallets or exchanges on public Wi-Fi.
- **Social Engineering** — Scammers manipulate you to reveal sensitive information.
- **Smart Contract Risks** — Bugs in smart contracts can lead to loss of funds.

COMMON CRYPTO SCAMS

Scam Type	How It Works	Red Flags	How to Protect Yourself	Real Examples
Ponzi / Pyramid Schemes	Promises high returns. Pays old investors with money from new investors.	• Guaranteed high returns • Referral bonuses • No real product	• If it sounds too good to be true, it probably is. • Check if there's a real product or revenue.	Bitconnect OneCoin
Phishing Attacks	Fake websites or emails trick you into giving your login, seed phrase or private keys.	• Urgent messages • Fake URLs • Poor grammar	• Never click on unknown links. • Always type the URL manually. • Verify the sender.	Fake MetaMask login Fake airdrop sites
Fake Giveaways & Airdrops	Scammers promise free crypto. Ask you to send crypto first or connect wallet to drain it.	• "Send crypto to receive more" • Unverified accounts	• Legit airdrops never ask for money. • Don't connect wallet randomly.	Fake Elon Musk giveaways
Rug Pulls	Project creators take investors' money and disappear.	• No team info • Locked liquidity • Anonymous developers	• Check team, audits and liquidity. • Use tools like TokenSniffer, RugDoc.	Squid Game Token Thodex
Pump & Dump	Scammers pump the price of a coin and dump their bags, leaving others with huge losses.	• Sudden price pump • Hype on social media • Low liquidity	• Don't FOMO. • Do your own research (DYOR).	Many low-cap tokens on Twitter/Telegram
Fake Support Scams	Scammers pretend to be support staff and ask for your keys or remote access.	• DMs from "support" • Ask for seed phrase • Remote access tools	• Real support never DM you first. • Never share your seed phrase.	Fake exchange support on Telegram
Malicious Smart Contracts	Bad contracts may steal approvals or drain your funds.	• Unknown contracts • Excessive permissions • No audit	• Review contract on Etherscan. • Revoke unused approvals.	Unlimited token approval drains

Wallet Security Checklist

- [x] Use hardware wallet for large holdings.
- [x] Store seed phrase offline.
- [x] Enable 2FA on all accounts.
- [x] Use strong, unique passwords.
- [x] Beware of phishing and fake links.
- [x] Keep software and wallets updated.
- [x] Never share your private keys.

How to Spot a Safe Project?

- Clear use case and product
- Verified team & advisors
- Active community
- Smart contract audited
- Transparency and road map
- Listed on trusted exchanges

What to Do If You Get Scammed?

1. Stop all further transactions.
2. Save all evidence (tx hash, chats, screenshots, emails).
3. Report to the platform/exchange.
4. Warn others in the community.
5. Learn and stay alert!

Useful Tools

- **Etherscan** – Check addresses and contracts
- **TokenSniffer / RugDoc** – Check token risks
- **Revoke.cash** – Revoke token approvals
- **HaveIBeenPwned** – Check data breaches

☆ Golden Rules

1. Never share your seed phrase.
2. Do your own research (DYOR).
3. Verify everything, trust slowly.
4. Security is in your hands.

Remember

Blockchain is secure, but humans are the weakest link.

Did You Know?

More than $1B+ is lost to scams, hacks and exploits every year!

Key Takeaway

Stay informed, stay cautious, and stay secure in crypto!

Secure today, safe tomorrow. Protect your crypto, protect your future!

13. GAS FEES + TRANSACTION FEES

What are Gas Fees?

- Gas fee is the fee you pay to process a transaction or execute a smart contract on a blockchain.
- You pay gas on networks like Ethereum, BNB Chain, Polygon, Avalanche, etc.
- It prevents spam and rewards validators/miners.



Gas Fee = Gas Price × Gas Used

1. **Gas Price (Gwei):** The price you are willing to pay per unit of gas.
2. **Gas Used:** The amount of gas required to complete the transaction.

- 1 Gwei = 0.000000001 ETH

Example

Gas Price : 30 Gwei
Gas Used : 21,000
Fee = 30 × 21,000 = 630,000 Gwei
= 0.00063 ETH

Higher gas price → Faster confirmation

What are Transaction Fees?

- Transaction fee is the general term for any fee paid to process a transaction on any blockchain.
- On Ethereum & similar chains, it is called gas fee.
- On Bitcoin, it is called miner fee.
- Fees can be in native token of the blockchain (e.g., ETH on Ethereum, BTC on Bitcoin).

ETH (Gas Fee) BTC (Miner Fee)

How Gas Fees Work (Step-by-Step)

1. You initiate a transaction
2. Transaction is broadcast to the network
3. You specify Gas Price & Gas Limit
4. Validators pick the transaction with higher fees first
5. Transaction is executed
6. Fee is paid to validators/miners
7. Transaction is confirmed & added to blockchain

Gas Price & Speed (Ethereum Example)

Priority Level	Gas Price (Gwei)	Speed	Use Case
Low	1 – 10	Slow (Minutes)	Non-urgent txns
Medium	10 – 30	Normal (1–2 mins)	Regular txns
High	30 – 100	Fast (15–60 sec)	Urgent txns
Very High	100+	Very Fast (Few sec)	Time-sensitive txns

Tip: You can check live gas prices on websites like Etherscan Gas Tracker.

Gas Limit vs Gas Used

- **Gas Limit:** The maximum amount of gas you allow for the transaction.
- **Gas Used:** The actual gas consumed.

If Gas Used ≤ Gas Limit

Transaction is successful.
You pay for the gas used.
Example:
Gas Limit = 100,000
Gas Used = 60,000
You pay for 60,000 gas only.

If Gas Used > Gas Limit

Transaction fails.
All gas is wasted.
Example:
Gas Limit = 60,000
Gas Used = 80,000
Transaction fails & gas is lost.

Gas Price on Popular Blockchains

Blockchain	Fee Name	Paid In	Notes
Ethereum	Gas Fee	ETH	Used for transactions & smart contracts
BNB Chain	Gas Fee	BNB	Lower fees compared to Ethereum
Polygon	Gas Fee	MATIC	Very low fees
Avalanche	Gas Fee	AVAX	Low fees, high speed
Bitcoin	Miner Fee	BTC	Based on size (bytes) & network demand
Solana	Transaction Fee	SOL	Very low fixed fees

What Affects Gas Fees?

- Network Demand (More users = Higher fees)
- Block Space Supply (Limited space per block)
- Gas Price (Set by you or suggested by wallet)
- Complexity of Transaction / Smart Contract (More computation = More gas used)

Where Do Fees Go?

Fees are paid to validators/miners for:

- Securing the network
- Processing & confirming your transaction

Tips to Save on Fees

- ☑ Use Layer 2 solutions (Arbitrum, Optimism, zkSync).
- ☑ Transact during off-peak hours.
- ☑ Use fee estimation tools in wallets.
- ☑ Avoid unnecessary on-chain actions.
- ☑ Batch transactions when possible.

Key Takeaway

- ☆ Gas Fee is the cost of doing anything on a blockchain.
- ☆ You pay for computation + validation.
- ☆ Higher fee → Faster transaction.
- ☆ Plan smart, pay less, transact more!

Did You Know?

On Ethereum, some DeFi swaps or NFTs may cost more in gas fee than the actual amount you are transacting!

Gas fees keep the blockchain secure and transactions running!
Understand, optimize and save.

14. WEB3 + DAOS + REAL-WORLD APPLICATIONS

What is Web3?

- Web3 is the decentralized version of the internet.
- Built on blockchain, crypto tokens, and smart contracts.
- Gives users ownership, privacy and control over their data and assets.
- No central authority. The users own the network.



Key Features of Web3

Decentralization
No single company or government controls the network.

Ownership
Users own their data, identity and assets through wallets.

Transparency
Transactions and rules are on-chain and open for everyone.

Permissionless
Anyone can build, access and participate.

Interoperability
Open protocols allow apps to work together.

Web3 vs Web2 (Quick Comparison)

Feature	Web2	Web3
Control	Centralized (Big Tech)	Decentralized
Data Ownership	Owned by companies	Owned by users
Data Privacy	Collected & sold	User-controlled
Trust	Need to trust intermediaries	Trust in code (not people)
Payments	Banks & gateways	Crypto & wallets
Censorship	Can be blocked	Hard to censor
Examples	Facebook, Google, Uber, Airbnb	Uniswak, Lens, Metamask, ENS

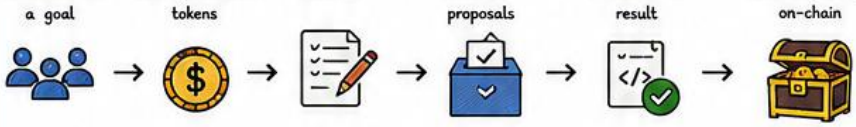
What is a DAO?

- DAO = Decentralized Autonomous Organization
- A community-run organization governed by code (smart contracts) and token holders.
- Decisions are made by voting, not by a central authority.
- Transparent, global and censorship-resistant.



How DAOs Work (Step-by-Step)

1. Community forms around a goal
2. Members buy or earn governance tokens
3. Proposals are created
4. Token holders vote on proposals
5. Smart contract executes the result
6. Treasury & operations run transparently on-chain



Code is law. Rules are transparent. Community is the boss.

Types of DAOs

Protocol DAO
Manages a DeFi or Web3 protocol (e.g., Uniswap DAO)

Investment DAO
Invests treasury in assets or startups (e.g., Floki DAO)

Social DAO
Built around communities and shared interests (e.g., Friends With Benefits)

Service DAO
Provides services in a decentralized way (e.g., Raid Guild)

Collector DAO
Collects and manages NFTs together (e.g., Flamingo DAO)

Popular DAO Examples

DAO	Focus	Governance Token	What it Does
Uniswap DAO	DeFi (DEX)	UNI	Governance of Uniswap protocol fees, upgrades, treasury.
MakerDAO	DeFi (Stablecoin)	MKR	Manages DAI stablecoin, risk parameters, upgrades.
Aave DAO	DeFi (Lending)	AAVE	Governance of Aave protocol, parameters & treasury.
Gitcoin DAO	Public Goods	GTC	Funds open-source & public good projects.
ENS DAO	Name Service	ENS	Governance of Ethereum Name Service.

Why DAOs Matter?

- ✓ Democratizes decision-making
- ✓ Aligns incentives with community
- ✓ Treasury managed transparently
- ✓ Global participation, 24/7
- ✓ Open-source & permissionless
- ✓ Reduces intermediaries & costs

Challenges of DAOs

- ✗ Low participation in voting
- ✗ Whale (rich) domination
- ✗ Complex governance
- ✗ Legal & regulatory uncertainty
- ✗ Smart contract vulnerabilities

Real-World Applications of Web3

Decentralized Finance (DeFi)
- Lending, borrowing, saving, trading
- No banks, open to all
- Examples: Aave, Compound, Uniswap

Decentralized Identity (DID)
- Own your identity and data
- No central authority
- Examples: ENS, Worldcoin, Ceramic

NFTs & Digital Ownership
- Own digital art, collectibles, land, tickets & more
- Verifiable scarcity & provenance
- Examples: OpenSea, Magic Eden

Gaming & Metaverse
- Play-to-earn games
- Virtual worlds owned by users
- Examples: Axie Infinity, The Sandbox

Decentralized Social Media
- User-owned content & platforms
- No censorship, no ads tracking
- Examples: Lens Protocol, Farcaster

Supply Chain & Traceability
- Track products on-chain
- Transparent & tamper-proof records
- Examples: VeChain, IBM Food Trust

Real Estate & RWAs (Real World Assets)
- Tokenize real estate, bonds, commodities
- Trade fractional ownership
- Examples: RealT, Centrifuge, Ondo

Web3 Stack (Basic Building Blocks)

- Blockchain – Decentralized ledger
- Smart Contracts – Code that runs on-chain
- Wallets – Store keys & interact with dApps
- dApps – Decentralized applications
- Tokens – Utility, governance, or assets
- Oracles – Bring real-world data on-chain

Benefits of Web3

- ☑ User ownership & control
- ☑ Privacy & data protection
- ☑ Transparency & auditability
- ☑ Global access & inclusion
- ☑ Innovation & new business models

Key Takeaway

Web3 empowers users.
DAOs empower communities.
Together, they are building a more open, fair, and user-owned internet.

Did You Know?

There are 10,000+ DAOs today managing billions of dollars in treasuries across DeFi, NFTs, gaming, and more!

Web3 is the future of the internet. DAOs are the future of organizations.
Own your future. Build. Participate. Decentralize.

15. CRYPTOCURRENCY INTERVIEW QUESTIONS + QUICK ANSWERS

1 What is Cryptocurrency?

Digital or virtual money that uses cryptography for security and works on decentralized blockchain networks.

2 What is Blockchain?

A decentralized, immutable ledger that records transactions across a network of computers in blocks linked together.

3 What is Decentralization?

It means no single authority or entity controls the network. Power is distributed among all participants.

4 What is the difference between Bitcoin and Altcoins?

Bitcoin was the first crypto and is mainly a store of value.
Altcoins are all other cryptocurrencies with different use cases.

5 What are Stablecoins?

Cryptocurrencies pegged to stable assets like USD to minimize price volatility.
Examples: USDT, USDC, DAI.

6 What are Tokens?

Tokens are digital assets built on existing blockchains, usually representing utility, security or governance.

7 What are NFTs?

Non-Fungible Tokens are unique digital assets representing ownership of items like art, music, collectibles, etc.

8 What is DeFi?

Decentralized Finance uses blockchain to offer financial services like lending, borrowing and trading without intermediaries.

9 What is DEX?

Decentralized Exchange allows users to trade crypto directly from their wallets without a central authority.

10 What is Mining?

Mining is the process of validating transactions and adding them to the blockchain using computational power.

11 What is Proof of Work (PoW)?

A consensus mechanism where miners solve complex puzzles to validate transactions and earn rewards. (e.g., Bitcoin)

12 What is Proof of Stake (PoS)?

A consensus mechanism where validators are chosen based on the number of coins they stake as collateral.

13 What are Gas Fees?

Fees paid to process transactions or execute smart contracts on blockchains like Ethereum.
Paid in the network's native token.

14 What is a Wallet?

A crypto wallet stores your private keys and allows you to send, receive and manage cryptocurrencies.

15 What is a Private Key?

A secret key that gives you access to your crypto. Never share it with anyone.

16 What is a Public Key / Address?

A public key is derived from your private key and is used to receive crypto. It's safe to share.

17 What is a Smart Contract?

Self-executing contracts with the terms of the agreement directly written in code on the blockchain.

18 What is 51% Attack?

When a single entity controls more than 50% of the network power and can manipulate transactions.

19 What is the difference between Cold Wallet and Hot Wallet?

Cold wallets are offline and more secure (e.g., hardware wallets).
Hot wallets are online and more convenient (e.g., mobile apps).

20 What is the purpose of Whitepaper?

A whitepaper explains the project's problem, solution, technology, tokenomics and roadmap in detail.

Bonus Tip for Interviews

- Understand the fundamentals deeply
- Know real-world use cases
- Stay updated with industry trends
- Be honest if you don't know
- Think out loud and reason clearly

BONUS QUICK FACTS

Bitcoin Genesis Block
Created on 3 Jan 2009 by Satoshi Nakamoto.

Total Bitcoin Supply
21 Milllion BTC is the maximum supply.

Fastest Blockchain
Solana can handle 65,000+ TPS (Transactions Per Second).

Largest DeFi TVL (2024)
Ethereum has the highest Total Value Locked in DeFi.

First NFT
"Quantum" by Kevin McCoy was one of the first NFTs (2014).

Most Popular Stablecoin
USDT (Tether) is the most used stablecoin.

Blockchain is a revolutionary technology. Keep learning, keep building!
Understand the WHY, not just the HOW.